UI/UX Design Strategy: ERP Item Management System

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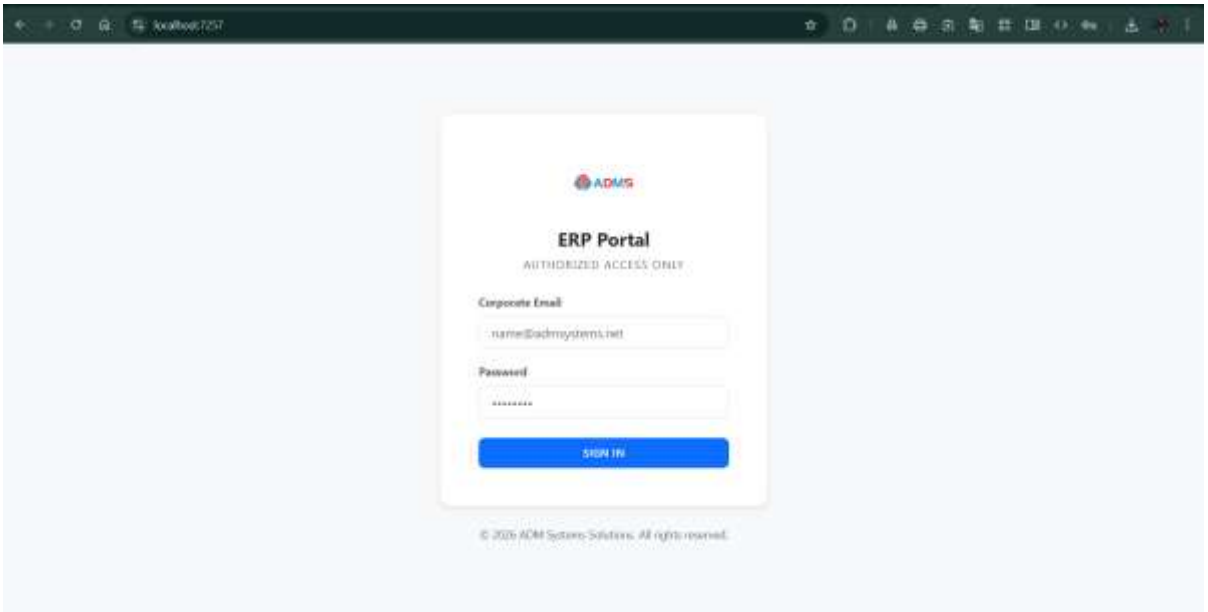
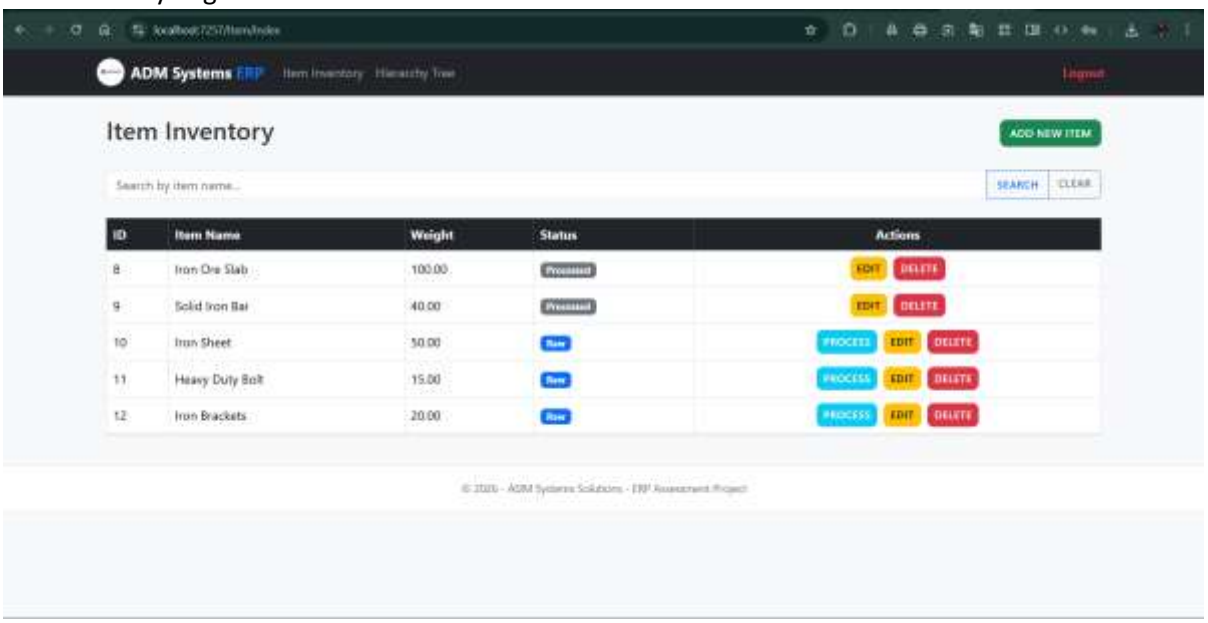
The UI/UX Layouts are:

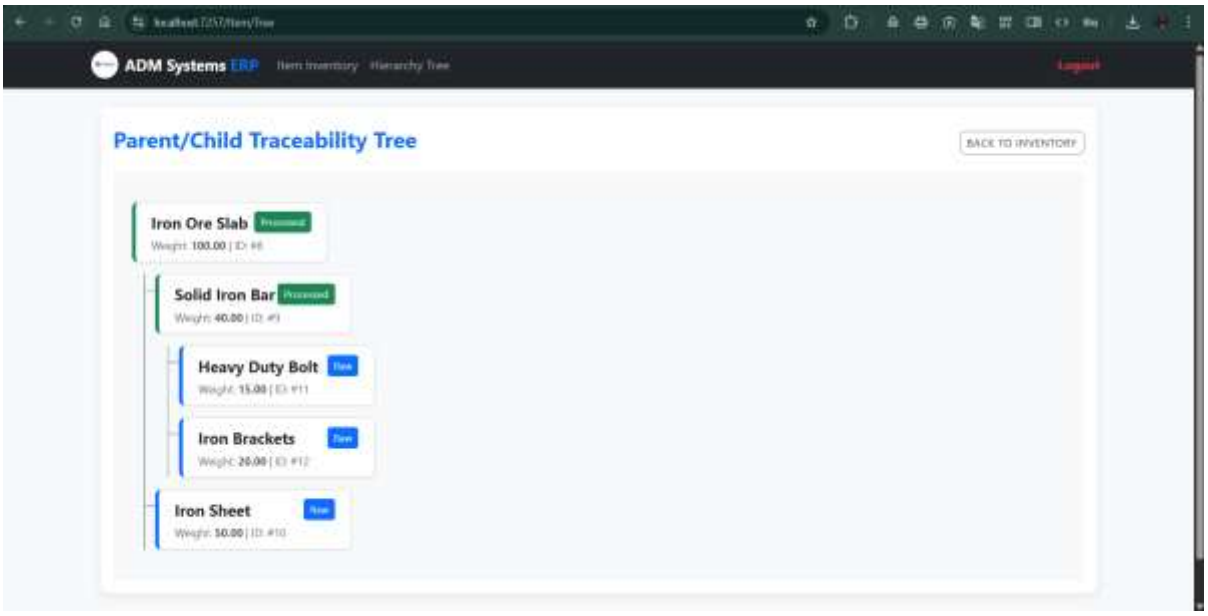
1) The Login Page:



2) The Inventory Page:



3) The Tree Hierarchy Page:



1. The Design Philosophy:

I designed this interface with a "low-fatigue" approach, using **Bootstrap 5** for a clean, professional look that users in an office or warehouse setting would find intuitive. I prioritized white space and clear borders to ensure that data remains the hero of the screen.

2. Visual Hierarchy and Information Scannability

In a system handling hundreds of items, a user needs to find information in seconds. I implemented several "visual cues" to help:

- **Color-Coded Status Badges:** I chose a high-contrast **Blue (Primary)** for "Raw" items and a neutral **Grey (Secondary)** for "Processed" items. This allows a user to scroll through a long list and instantly identify which items still require action without needing to read the text in every row.
- **Conditional Interface Logic:** To reduce "UI Clutter," I wrote logic that hides the "Process" button once an item has already been broken down. This prevents accidental duplicate entries and keeps the "Actions" column clean.

3. Dynamic Workflow for Real-World Scenarios

In actual manufacturing or inventory management, a single raw material doesn't always break down into a fixed number of parts.

- **The "Add Row" Feature:** I built the processing screen with a dynamic JavaScript-powered table. This allows the user to add as many output items as they need on the fly. It adapts to the worker’s reality, rather than forcing the worker to follow a rigid, pre-set form.

- **Persistent Context:** On the processing screen, I kept the "Parent Item" details visible at the top. This ensures the user always knows exactly which material they are currently accounting for.

4. Recursive Traceability (The Tree View)

The most complex part of the system is the Parent/Child relationship.

- **Logical Mapping:** Instead of a confusing nested list, I built a visual "Tree" using custom CSS lines.
- **Indentation Strategy:** Each "Child" is indented under its "Parent," mimicking a file-folder structure.
- **Node Accents:** Each box in the tree has a coloured left border (Green for finished, Blue for raw). This gives a sense of "health" or "completion" to the entire production chain at a single glance.

5. Safety and Error Prevention

- **Delete Confirmations:** I added a simple JavaScript prompt for the Delete action. In a professional setting, a single mis-click shouldn't result in lost data.
- **Mobile-Ready Navigation:** I used a standard "Hamburger" menu for mobile views, ensuring that a supervisor checking inventory on a tablet or phone has the same seamless experience as someone at a desktop.